

Lab 4: VibeCoding (Lunar Lander) Chat History

LLM: Claude (Claude Code CLI, Opus 5.5)

USER

read @README.md and /home/aditya/Documents/SE/SE-LABS/Lab-04/Lab_4_VibeCoding_Student_handout.pdf, understand the requirements and complete all the given tasks make sure you also create a pdf of this chat, rn i have cloned the repo from SETAPESU26, delete the git tracking for this and instead create a new folder under my https://github.com/8bitGuitar/SE-Labs named Lab-04 for the same, I have already recorded the "before" video, just finsih the tasks and ill record the "after" one myself too

CLAUDE

I'll start by reading the handout and the game code.

Tool call: Bash

List lab files and extract handout text

```
$ cd /home/aditya/Documents/SE/SE-LABS/Lab-04 && ls -la && ls -la 32_lunarLander && (pdftotext -layout Lab_4_VibeCoding_Student_handout.pdf - 2>/dev/null || echo "no pdftotext")
```

Tool call: Read

/home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py

Result

```
total 84
drwxr-xr-x 3 aditya aditya 4096 Sep 30 08:30 .
drwxr-xr-x 7 aditya aditya 4096 Sep 30 08:30 ..
drwxr-xr-x 4 aditya aditya 4096 Sep 30 08:42 32_lunarLander
-rw-r--r-- 1 aditya aditya 71355 Sep 30 08:28 Lab_4_VibeCoding_Student_handout.pdf
total 32
drwxr-xr-x 4 aditya aditya 4096 Sep 30 08:42 .
drwxr-xr-x 3 aditya aditya 4096 Sep 30 08:30 ..
-rw-r--r-- 1 aditya aditya 7191 Sep 30 08:31 game.py
drwxr-xr-x 7 aditya aditya 4096 Sep 30 08:43 .git
-rw-r--r-- 1 aditya aditya 13 Sep 30 08:31 .gitignore
drwxr-xr-x 2 aditya aditya 4096 Sep 30 08:42 __pycache__
-rw-r--r-- 1 aditya aditya 4028 Sep 30 08:31 README.md
Lab 4: VibeCoding
Objective
Use Vibe Coding tools to fix the broken code and add features to the same. Learn how
to prompt to get the job done in under three to four attempts. Clone a repo assigned
to you, make changes and push to your repo in your personal account.

THIS IS AN INDIVIDUAL ASSIGNMENT. WORK INDIVIDUALLY!!

Duration
60 minutes

Software Requirements
● Access to Vibe Coding tool. (ChatGPT, Gemini, Claude etc)

Instructions:
1. Go to the repo assigned to you
```

2. Go through the ReadMe file to understand the deliverables
3. Clone or Fork the repo
4. Run the Python based code (A game/ A Puzzle)
5. Record 10 seconds of video before making any changes.
6. Write prompts and fix the broken code
7. Write prompts to add all the features listed in the ReadMe file of the repo one by one.
8. Record another 10 seconds of video again after making all the changes.
9. Push the code to your repo.
10. Do not raise PR to the main repo (SETAPESU26)

Deliverables:

Upload the following to your Lab repo under Lab-4 folder,

- a. Videos (before and after)
- b. Updated Code
- c. Chat history exported as a doc/pdf

Shell cwd was reset to /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander

Result

```
1     import math
2     import random
3     import pygame
4
5     WIDTH, HEIGHT = 800, 600
6     STEP = 20
7     THRUST, ROTATE_SPEED, BURN_RATE, FUEL_MAX = 60.0, 2.4, 22.0, 300.0
8     MAX_SPEED_X, MAX_SPEED_Y, MAX_ANGLE = 25.0, 40.0, 0.25
9     FOOT = 12
10
11
12     def ship_color(fuel_ratio):
13         """Return an (r, g, b) hull colour for the given fuel ratio (1.0 = full), or None for the
14         default."""
15         pass
16
17     def on_landing(score):
18         """Called after a successful landing with the points just earned; add fireworks or bonuses
19         here."""
20         pass
21
22     def bonus_life_threshold():
23         """Return a score value at which the player earns an extra life, or None to disable bonus
24         lives."""
25         pass
26
27     def make_terrain():
28         heights, y = [], random.randint(430, 520)
29         for _ in range(WIDTH // STEP + 1):
30             y = max(380, min(560, y + random.randint(-32, 32)))
31             heights.append(y)
32         pads = []
33         for start, length, mult in ((random.randint(2, 12), 4, 1), (random.randint(20, 35), 3, 3)):
34             for i in range(start, start + length):
35                 heights[i] = heights[start]
36             pads.append((start * STEP, (start + length - 1) * STEP, heights[start], mult))
37         return heights, pads
38
```

```

39
40     def ground_y(heights, x):
41         x = max(0, min(WIDTH - 1, x))
42         i = int(x // STEP)
43         t = (x - i * STEP) / STEP
44         return heights[i] * (1 - t) + heights[i + 1] * t
45
46
47     def wrap_angle(angle):
48         return (angle + math.pi) % math.tau - math.pi
49
50
51     class Game:
52         def __init__(self):
53             self.font = pygame.font.Font(None, 26)
54             self.reset()
55
56         def reset(self):
57             self.level, self.score, self.lives = 1, 0, 3
58             self.bonus_awarded = 0
59             self.new_round()
60
61     ... (125 more lines truncated)

```

CLAUDE

Requirements are clear: fix 1 bug, implement 3 hooks, push to `8bitGuitar/SE-Labs` under `Lab-04`, and include a PDF of this chat. Let me check the target repo and tooling before editing.

Tool call: Bash

Check remote, gh auth, target repo contents, tooling

```

$ git remote -v && cat .gitignore && gh auth status 2>&1 | head -5; gh api repos/8bitGuitar/SE-
Labs/contents --jq '.[].name' 2>&1; python -c "import pygame; print(pygame.version.ver)"; which pandoc
wkhtmltopdf chromium google-chrome-stable weasyprint 2>/dev/null; python -c "import reportlab;
print('reportlab ok')" 2>&1 | tail -1

```

Result

```

origin https://github.com/SETAPESU26/32_lunarLander.git (fetch)
origin https://github.com/SETAPESU26/32_lunarLander.git (push)
__pycache__/
github.com
  ✓ Logged in to github.com account Aditya-Vats-Ensomate (keyring)
  - Active account: true
  - Git operations protocol: https
  - Token: gho_*****
.gitignore
Lab-02
Lab-03
Lab-1
README.md
<frozen importlib._bootstrap>:491: RuntimeWarning: Your system is avx2 capable but pygame was not built
with support for it. The performance of some of your blits could be adversely affected. Consider
enabling compile time detection with environment variables like PYGAME_DETECT_AVX2=1 if you are
compiling without cross compilation.
/home/aditya/.local/lib/python3.14/site-packages/pygame/pkgdata.py:25: UserWarning: pkg_resources is
deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources
package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to
Setuptools<81.
  from pkg_resources import resource_stream, resource_exists
pygame 2.6.1 (SDL 2.32.72, Python 3.14.7)
Hello from the pygame community. https://www.pygame.org/contribute.html

```

```
2.6.1
pandoc not found
wkhtmltopdf not found
chromium not found
/bin/google-chrome-stable
weasyprint not found
ModuleNotFoundError: No module named 'reportlab'
```

CLAUDE

Note: `gh` is logged in as `Aditya-Vats-Ensomate`, not `8bitGuitar` — I'll check push access when I get there. Let me see how previous labs are laid out in SE-Labs.

Tool call: Bash

Inspect Lab-03 layout and repo permissions

```
$ gh api repos/8bitGuitar/SE-Labs/contents/Lab-03 --jq '.[ ] | "\(.type) \(.name)"; echo ---; gh api repos/8bitGuitar/SE-Labs --jq '{default_branch, perms: .permissions}'
```

Result

```
file Lab3_Architecture_Justification.pdf
file Lab3_Component_Diagram.pdf
---
{"default_branch":"main","perms":
{"admin":false,"maintain":false,"pull":true,"push":false,"triage":false}}
```

CLAUDE

The `gh` account has no push access to SE-Labs. Let me check whether an SSH key for 8bitGuitar is set up.

Tool call: Bash

Check which GitHub account the SSH key authenticates as

```
$ ssh -T -o BatchMode=yes -o ConnectTimeout=10 git@github.com 2>&1; ls ~/.ssh 2>/dev/null; git config -global credential.helper
```

Result

```
Host key verification failed.
known_hosts
known_hosts.old
/usr/lib/git-core/git-credential-libsecret
```

Tool call: Edit

/home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py

```
if pad and abs(self.vel.y) <= MAX_SPEED_Y and abs(angle) <= MAX_ANGLE:
```

```
    if (pad and abs(self.vel.x) <= MAX_SPEED_X and abs(self.vel.y) <= MAX_SPEED_Y
        and abs(angle) <= MAX_ANGLE):
```

Tool call: Edit

/home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py

```
elif abs(angle) > MAX_ANGLE:
    reason = "bad angle"
else:
```

```
elif abs(angle) > MAX_ANGLE:
    reason = "bad angle"
```

```
elif abs(self.vel.x) > MAX_SPEED_X:
    reason = "drifting sideways"
else:
```

Result

The file /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py has been updated successfully.
(file state is current in your context – no need to Read it back)

Result

The file /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py has been updated successfully.
(file state is current in your context – no need to Read it back)

Tool call: Edit

/home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py

```
def ship_color(fuel_ratio):
    """Return an (r, g, b) hull colour for the given fuel ratio (1.0 = full), or None for the
    default."""
    pass

def on_landing(score):
    """Called after a successful landing with the points just earned; add fireworks or bonuses here."""
    pass

def bonus_life_threshold():
    """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
    pass
```

```
FULL_COLOR, EMPTY_COLOR = (230, 230, 240), (255, 50, 40)
FIREWORK_COLORS = [(255, 90, 90), (255, 210, 80), (120, 220, 255), (180, 120, 255), (120, 255, 150)]
particles = [] # each: [x, y, vx, vy, life, color]

def ship_color(fuel_ratio):
    """Return an (r, g, b) hull colour for the given fuel ratio (1.0 = full), or None for the
    default."""
    t = max(0.0, min(1.0, fuel_ratio))
    return tuple(int(e + (f - e) * t) for f, e in zip(FULL_COLOR, EMPTY_COLOR))

def on_landing(score):
    """Called after a successful landing with the points just earned; add fireworks or bonuses here."""
    for _ in range(min(8, 1 + score // 200)): # bigger payouts (x3 pad) get more bursts
        cx, cy = random.uniform(120, WIDTH - 120), random.uniform(90, 260)
        color = random.choice(FIREWORK_COLORS)
        delay = random.uniform(0.0, 0.8)
        for _ in range(40):
            angle, speed = random.uniform(0, math.tau), random.uniform(40, 170)
            particles.append([cx, cy, math.cos(angle) * speed, math.sin(angle) * speed,
                             1.4 + delay, color, delay])

def update_particles(dt):
    for p in particles:
        p[4] -= dt
        if p[6] > 0: # burst still waiting to go off
            p[6] -= dt
            continue
        p[2] *= 0.98
        p[3] = p[3] * 0.98 + 60 * dt
        p[0] += p[2] * dt
```

```

        p[1] += p[3] * dt
        particles[:] = [p for p in particles if p[4] > 0]

def draw_particles(screen):
    for x, y, _, _, life, color, delay in particles:
        if delay <= 0:
            fade = min(1.0, life / 0.6)
            pygame.draw.circle(screen, tuple(int(c * fade) for c in color), (x, y), 2)

def bonus_life_threshold():
    """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
    return 1500

```

Tool call: Edit

/home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py

```
screen.blit(label, label.get_rect(center=(WIDTH // 2, HEIGHT // 3)))
```

```

        screen.blit(label, label.get_rect(center=(WIDTH // 2, HEIGHT // 3)))
        draw_particles(screen)

```

Tool call: Edit

/home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py

```

game.update(dt, pygame.key.get_pressed())
game.draw(screen)

```

```

game.update(dt, pygame.key.get_pressed())
update_particles(dt)
game.draw(screen)

```

Result

The file /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py has been updated successfully. (file state is current in your context – no need to Read it back)

Result

The file /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py has been updated successfully. (file state is current in your context – no need to Read it back)

Result

The file /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py has been updated successfully. (file state is current in your context – no need to Read it back)

Tool call: Bash

Fix particle layout comment

```
$ sed -i 's/particles = \[\] # each: \[x, y, vx, vy, life, color\]/particles = [] # each: [x, y, vx, vy, life, color, delay]/' game.py && grep -n "particles = \[\]" game.py
```

Result

```
14:particles = [] # each: [x, y, vx, vy, life, color, delay]
```

CLAUDE

Now a headless test that exercises each task: the sideways-slam case, the other landing rules, the fuel colours, the bonus life, and the fireworks lifecycle, plus a rendered frame I can look at.

Tool call: Bash

```
$ SDL_VIDEODRIVER=dummy SDL_AUDIODRIVER=dummy python - <<'EOF' 2>&1 | grep -v -i -E
"avx2|pkg_resources|resource_stream|pygame community|^pygame "
import pygame, game as g
pygame.init()
screen = pygame.display.set_mode((g.WIDTH, g.HEIGHT))
G = g.Game()

def land(vx, vy, angle, on_pad=True):
    G.reset(); G.state = "fly"
    x1, x2, y, m = G.pads[1]
    G.pos = pygame.Vector2((x1 + x2) / 2 if on_pad else (x1 - 60) % g.WIDTH, y - g.FOOT)
    G.vel = pygame.Vector2(vx, vy); G.angle = angle
    G.touchdown()
    return G.state, G.message

print("Task1 sideways slam :", land(80, 20, 0))
print("Task1 gentle landing :", land(10, 20, 0))
print("Task1 too fast (Vy) :", land(0, 60, 0))
print("Task1 bad angle :", land(0, 10, 0.5))
print("Task1 edge Vx=25 :", land(25, 20, 0)[0], "| Vx=-25.1:", land(-25.1, 20, 0)[0])

print("Task2 colors full/half/empty:", g.ship_color(1.0), g.ship_color(0.5), g.ship_color(0.0),
g.ship_color(-0.2))

g.particles.clear(); g.on_landing(400); n1 = len(g.particles)
g.particles.clear(); g.on_landing(1200); n3 = len(g.particles)
print(f"Task3 particles x1-pad(400)={n1} x3-pad(1200)={n3}")
for _ in range(200): g.update_particles(1/60)
print("Task3 particles after ~3.3s:", len(g.particles))

G.reset(); G.score = 1600; G.update(0.016, pygame.key.get_pressed())
print("Task4 lives at 1600:", G.lives, "| again same score:", (G.update(0.016,
pygame.key.get_pressed()), G.lives)[1])
G.score = 3100; G.update(0.016, pygame.key.get_pressed()); print("Task4 lives at 3100:", G.lives)

# render a frame mid-fireworks for visual check
land(5, 15, 0); G.fuel = 60
for _ in range(50): g.update_particles(1/60)
G.draw(screen); pygame.image.save(screen, "/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-
32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/frame.png")
print("frame saved")
EOF
```

Result

```
Task1 sideways slam : ('crashed', 'Crashed: drifting sideways! Space = retry')
Task1 gentle landing : ('landed', 'Perfect landing! +1200 (Space = next level)')
Task1 too fast (Vy) : ('crashed', 'Crashed: too fast! Space = retry')
Task1 bad angle : ('crashed', 'Crashed: bad angle! Space = retry')
Task1 edge Vx=25 : landed | Vx=-25.1: crashed
Task2 colors full/half/empty: (230, 230, 240) (242, 140, 140) (255, 50, 40) (255, 50, 40)
Task3 particles x1-pad(400)=120 x3-pad(1200)=280
Task3 particles after ~3.3s: 0
Task4 lives at 1600: 4 | again same score: 4
Task4 lives at 3100: 5
Traceback (most recent call last):
  File "<stdin>", line 35, in <module>
NotImplementedError: saving images of extended format is not available
```

CLAUDE

Every check passes. The failure at the end is only because this pygame build can't save PNG, so I'll save the frame as BMP and look at it.

Tool call: Bash

Render a fireworks frame to an image

```
$ S=/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad && SDL_VIDEODRIVER=dummy python - <<EOF 2>/dev/null
import pygame, game as g
pygame.init(); screen = pygame.display.set_mode((g.WIDTH, g.HEIGHT)); G = g.Game()
x1, x2, y, m = G.pads[1]
G.pos = pygame.Vector2((x1+x2)/2, y-g.FOOT); G.vel = pygame.Vector2(5, 15); G.touchdown(); G.fuel = 60
for _ in range(50): g.update_particles(1/60)
G.draw(screen); pygame.image.save(screen, "$S/frame.bmp")
EOF
magick "$S/frame.bmp" "$S/frame.png" 2>/dev/null || convert "$S/frame.bmp" "$S/frame.png"; ls $S
```

Result

```
pygame 2.6.1 (SDL 2.32.72, Python 3.14.7)
Hello from the pygame community. https://www.pygame.org/contribute.html
frame.bmp
frame.png
```

Tool call: Read

/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/frame.png

Result

[image output]

Tool call: Bash

Check whether SE-LABS is a git clone

```
$ cd /home/aditya/Documents/SE/SE-LABS && ls -la && git -C /home/aditya/Documents/SE/SE-LABS rev-parse --show-toplevel 2>&1; git -C /home/aditya/Documents/SE/SE-LABS remote -v 2>&1; git -C /home/aditya/Documents/SE/SE-LABS status --short 2>&1 | head
```

Result

```
total 36
drwxr-xr-x 7 aditya aditya 4096 Sep 30 08:30 .
drwxr-xr-x 6 aditya aditya 4096 Sep 26 17:17 ..
drwxr-xr-x 9 aditya aditya 4096 Sep 30 08:42 .git
-rw-r--r-- 1 aditya aditya 200 Aug 27 08:52 .gitignore
drwxr-xr-x 2 aditya aditya 4096 Sep 7 14:59 Lab-02
drwxr-xr-x 2 aditya aditya 4096 Sep 10 23:02 Lab-03
drwxr-xr-x 3 aditya aditya 4096 Sep 30 08:30 Lab-04
drwxr-xr-x 2 aditya aditya 4096 Aug 27 11:35 Lab-1
-rw-r--r-- 1 aditya aditya 109 Aug 27 11:51 README.md
/home/aditya/Documents/SE/SE-LABS
origin https://github.com/8bitGuitar/SE-Labs.git (fetch)
origin https://github.com/8bitGuitar/SE-Labs.git (push)
?? Lab-04/
Shell cwd was reset to /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander
```

Tool call: Bash

Check gitignore, sync state, and push auth via dry run

```
$ cd /home/aditya/Documents/SE/SE-LABS && cat .gitignore && echo --- && git fetch origin 2>&1 && git status -sb | head -3 && git push --dry-run origin main 2>&1 | tail -3
```

Result

```
# Secrets and environment
.env
.env.*
*.pem
*.key
credentials.json

# OS and editor
.DS_Store
Thumbs.db
*.swp
*~

# Dependencies and build output
node_modules/
__pycache__/
*.pyc
dist/
build/
target/
---
## main...origin/main
?? Lab-04/
Everything up-to-date
Shell cwd was reset to /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander
```

CLAUDE

Push auth works through your stored git credentials. Now I'll remove the SETAPESU26 git tracking from the lab folder.

Tool call: Bash

Remove nested SETAPESU26 git repo and pycache

```
$ git remote get-url origin && git status --short && rm -rf .git __pycache__ && ls -la
```

Result

```
https://github.com/SETAPESU26/32_lunarLander.git
M game.py
total 28
drwxr-xr-x 2 aditya aditya 4096 Sep 30 08:46 .
drwxr-xr-x 3 aditya aditya 4096 Sep 30 08:30 ..
-rw-r--r-- 1 aditya aditya 8788 Sep 30 08:45 game.py
-rw-r--r-- 1 aditya aditya  13 Sep 30 08:31 .gitignore
-rw-r--r-- 1 aditya aditya 4028 Sep 30 08:31 README.md
```

CLAUDE

Now the chat PDF. Claude Code saves this session as a JSONL transcript, so I'll write a small exporter that renders it to HTML and prints it to PDF with headless Chrome.

Tool call: Bash

Locate and inspect session transcript structure

```
$ ls -la ~/.claude/projects/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/ && head -c 1500
~/.claude/projects/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-
1630c17fdf4a.jsonl; echo; python -c "
import json
```

```

from collections import Counter
c=Counter()
for l in open('/home/aditya/.claude/projects/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-
lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a.jsonl'):
    d=json.loads(l); m=d.get('message') or {}
    cont=m.get('content')
    kinds = [x.get('type') for x in cont] if isinstance(cont,list) else [type(cont).__name__]
    c[(d.get('type'), tuple(kinds))]+=1
for k,v in c.items(): print(v,k)
"

```

Result

```

total 804
drwxr-xr-x  3 aditya aditya   4096 Sep 30 08:43 .
drwxr-xr-x 13 aditya aditya   4096 Sep 30 08:39 ..
-rw-----  1 aditya aditya 332103 Sep 30 08:42 46d4249a-325c-46d4-8d51-ff4679de4597.jsonl
-rw-----  1 aditya aditya 473760 Sep 30 08:46 b52081dc-61a0-4d82-bf2d-1630c17fdf4a.jsonl
drwxr-xr-x  2 aditya aditya   4096 Sep 30 08:39 memory
{"type": "mode", "mode": "normal", "sessionId": "b52081dc-61a0-4d82-bf2d-1630c17fdf4a"}
{"type": "permission-mode", "permissionMode": "auto", "sessionId": "b52081dc-61a0-4d82-bf2d-1630c17fdf4a"}
{"type": "atis-latch", "atis": "", "sessionId": "b52081dc-61a0-4d82-bf2d-1630c17fdf4a"}
{"type": "file-history-snapshot", "messageId": "18e3739f-162e-419d-9cef-0330841e00a4", "snapshot":
{"messageId": "18e3739f-162e-419d-9cef-0330841e00a4", "trackedFileBackups": {}, "timestamp": "2026-09-
30T03:13:49.581Z"}, "isSnapshotUpdate": false}
{"parentUuid": null, "isSidechain": false, "promptId": "1efd6340-9177-45c6-8dc3-
d8104b3541e8", "type": "user", "message": {"role": "user", "content": " read @README.md and
/home/aditya/Documents/SE/SE-LABS/Lab-04/Lab_4_VibeCoding_Student_handout.pdf, understand the
requirements and complete all the given tasks make sure you also create a pdf of this chat, rn i have
cloned the repo from SETAPESU26, delete the git tracking for this and instead create a new folder under
my https://github.com/8bitGuitar/SE-Labs named Lab-04 for the same, I have already recorded the
\"before\" video, just finsih the tasks and ill record the \"after\" one myself too"}, "uuid": "18e3739f-
162e-419d-9cef-0330841e00a4", "timestamp": "2026-09-30T03:13:49.517Z", "permissionMode": "auto", "origin":
{"kind": "human"}, "promptSource": "typed", "turnOrigin": "human", "userType": "external", "entrypoint": "cli", "
cwd": "/home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander", "sessionId": "b52081dc-61a0-4d82-bf2d-
1630c17fdf4a", "version": "2.1.283"
7 ('mode', ('NoneType',))
7 ('permission-mode', ('NoneType',))
7 ('atis-latch', ('NoneType',))
1 ('file-history-snapshot', ('NoneType',))
1 ('user', ('str',))
36 ('attachment', ('NoneType',))
6 ('last-prompt', ('NoneType',))
6 ('ai-title', ('NoneType',))
15 ('assistant', ('thinking',))
7 ('assistant', ('text',))
17 ('assistant', ('tool_use',))
17 ('user', ('tool_result',))
1 ('file-history-delta', ('NoneType',))

```

Tool call: Write

/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/export_chat.py

```

"""Render a Claude Code session transcript (.jsonl) to a printable HTML page."""
import html
import json
import sys

src, out = sys.argv[1], sys.argv[2]
MAX_LINES = 60

def esc(s):

```

```

    return html.escape(str(s))

def clip(text):
    lines = str(text).splitlines()
    if len(lines) > MAX_LINES:
        lines = lines[:MAX_LINES] + [f"... ({len(lines) - MAX_LINES} more lines truncated)"]
    return "\n".join(lines)

def result_text(content):
    if isinstance(content, str):
        return content
    parts = []
    for c in content or []:
        if c.get("type") == "text":
            parts.append(c["text"])
        elif c.get("type") == "image":
            parts.append("[image output]")
    return "\n".join(parts)

def tool_call(block):
    name, inp = block["name"], block.get("input", {})
    if name == "Bash":
        body = f"<div class=desc>{esc(inp.get('description', ''))}</div><pre>${esc(clip(inp.get('command', '')))}</pre>"
    elif name == "Edit":
        body = (f"<div class=desc>{esc(inp.get('file_path'))}</div>"
                f"<pre class=old>{esc(clip(inp.get('old_string', '')))}</pre>"
                f"<pre class=new>{esc(clip(inp.get('new_string', '')))}</pre>")
    elif name == "Write":
        body = f"<div class=desc>{esc(inp.get('file_path'))}</div><pre class=new>{esc(clip(inp.get('content', '')))}</pre>"
    elif name == "Read":
        body = f"<div class=desc>{esc(inp.get('file_path'))}</div>"
    else:
        body = f"<pre>{esc(clip(json.dumps(inp, indent=2)))}</pre>"
    return f"<div class=tool><div class=tname>Tool call: {esc(name)}</div>{body}</div>"

rows = []
for line in open(src, encoding="utf-8"):
    d = json.loads(line)
    if d.get("type") not in ("user", "assistant") or d.get("isSidechain"):
        continue
    content = d["message"]["content"]
    if d["type"] == "user" and isinstance(content, str):
        rows.append(f"<div class=msg user><div class=who>User</div><div class=text>{esc(content)}</div></div>")
        continue
    for block in content:
        t = block.get("type")
        ... (26 more lines truncated)

```

Result

File created successfully at: /tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/export_chat.py (file state is current in your context – no need to Read it back)

Tool call: Bash

Test-render chat transcript to PDF with headless Chrome

```
$ S=/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad && python $S/export_chat.py ~/.claude/projects/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a.jsonl $S/chat.html && google-chrome-stable --headless=new --disable-gpu --no-pdf-header-footer --print-to-pdf=$S/chat_test.pdf file://$S/chat.html 2>&1 | tail -2; pdftinfo $S/chat_test.pdf | grep Pages && pdftoppm -png -r 60 -f 1 -l 1 $S/chat_test.pdf $S/p
```

Result

```
136660 bytes written to file /tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/chat_test.pdf
Pages: 11
```

USER (SENT MID-TASK)

create 1 commit for each task

CLAUDE

Got it, one commit per task. Your original code will go in first as a baseline commit so each task commit shows only its own diff. First, a quick look at the test PDF render:

Tool call: Read

```
/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/p-01.png
```

Result

[image output]

CLAUDE

The PDF renders cleanly. The nested `.git` is gone, so I'll pull the untouched original game.py` from the SETAPESU26 repo to use as the baseline.`

Tool call: Bash

Fetch original game.py and back up final version

```
$ S=/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad && gh api repos/SETAPESU26/32_lunarLander/contents/game.py -H "Accept: application/vnd.github.raw" > $S/orig_game.py && wc -c $S/orig_game.py && grep -n "abs(self.vel.y) <= MAX_SPEED_Y and" $S/orig_game.py && cp /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander/game.py $S/final_game.py
```

Result

```
7191 /tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/orig_game.py
77:         if pad and abs(self.vel.y) <= MAX_SPEED_Y and abs(angle) <= MAX_ANGLE:
```

CLAUDE

It matches the original byte count (7191). Next I'll build each stage by applying the task's edits in order, and check that the last stage is byte-identical to the tested final file.

Tool call: Bash

Build per-task file stages and verify final matches

```
$ S=/tmp/closure-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad && cd $S && python - <<'EOF'
final = open("final_game.py").read()
src = open("orig_game.py").read()

def seg(start, end):
    """Slice of the final file from `start` up to (not including) `end`."""
    i = final.index(start); return final[i:final.index(end, i)]

def rep(s, old, new):
    assert s.count(old) == 1, old; return s.replace(old, new)

# Task 1: touchdown validation
s1 = rep(src, "            if pad and abs(self.vel.y) <= MAX_SPEED_Y and abs(angle) <= MAX_ANGLE:\n",
          "            if (pad and abs(self.vel.x) <= MAX_SPEED_X and abs(self.vel.y) <= MAX_SPEED_Y\nand abs(angle) <= MAX_ANGLE):\n")
s1 = rep(s1, '                reason = "bad angle"\n                    else:',
          '                reason = "bad angle"\n                    elif abs(self.vel.x) > MAX_SPEED_X:\nreason = "drifting sideways"\n                    else:')

# Task 2: ship_color
s2 = rep(s1, "\n\ndef ship_color(fuel_ratio):", "\nFULL_COLOR, EMPTY_COLOR = (230, 230, 240), (255, 50, 40)\n\n\ndef ship_color(fuel_ratio):")
s2 = rep(s2, """"Return an (r, g, b) hull colour for the given fuel ratio (1.0 = full), or None for the default."""\n    pass',
          seg('""""Return an (r, g, b) hull colour', "\n\n\ndef on_landing"))

# Task 3: on_landing fireworks
s3 = rep(s2, "(255, 50, 40)\n", "(255, 50, 40)\n" + seg("FIREWORK_COLORS", "\n\n\ndef ship_color") + "\n")
s3 = rep(s3, """"Called after a successful landing with the points just earned; add fireworks or bonuses here."""\n    pass',
          seg('""""Called after a successful landing', "\n\n\ndef bonus_life_threshold"))
s3 = rep(s3, "(WIDTH // 2, HEIGHT // 3))\n", "(WIDTH // 2, HEIGHT // 3))\n
draw_particles(screen)\n")
s3 = rep(s3, "        game.update(dt, pygame.key.get_pressed())\n", "        game.update(dt,
pygame.key.get_pressed())\n        update_particles(dt)\n")

# Task 4: bonus life threshold
s4 = rep(s3, """"Return a score value at which the player earns an extra life, or None to disable bonus lives."""\n    pass',
          '""""Return a score value at which the player earns an extra life, or None to disable bonus lives."""\n    return 1500')

assert s4 == final, "stage 4 does not reproduce the final file"
for n, s in enumerate([src, s1, s2, s3, s4]):
    open(f"stage{n}.py", "w").write(s)
    compile(s, f"stage{n}", "exec")
print("all stages built; stage4 == final")
EOF
diff stage0.py stage1.py | head -20; echo ===; diff stage2.py stage3.py | head -8
```

Result

```
Exit code 1
Traceback (most recent call last):
  File "<stdin>", line 33, in <module>
AssertionError: stage 4 does not reproduce the final file
diff: stage0.py: No such file or directory
diff: stage1.py: No such file or directory
(eval):40: == not found
```

Tool call: Bash

Inspect top of final file to find stage mismatch

```
$ S=/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad && cd $S && sed -n 1,20p final_game.py | cat -A | head -20
```

Result

```
import math$
import random$
import pygame$
$
WIDTH, HEIGHT = 800, 600$
STEP = 20$
THRUST, ROTATE_SPEED, BURN_RATE, FUEL_MAX = 60.0, 2.4, 22.0, 300.0$
MAX_SPEED_X, MAX_SPEED_Y, MAX_ANGLE = 25.0, 40.0, 0.25$
FOOT = 12$
$
$
FULL_COLOR, EMPTY_COLOR = (230, 230, 240), (255, 50, 40)$
FIREWORK_COLORS = [(255, 90, 90), (255, 210, 80), (120, 220, 255), (180, 120, 255), (120, 255, 150)]$
particles = [] # each: [x, y, vx, vy, life, color, delay]$
$
$
def ship_color(fuel_ratio):$
    """Return an (r, g, b) hull colour for the given fuel ratio (1.0 = full), or None for the
    default."""$
    t = max(0.0, min(1.0, fuel_ratio))$
    return tuple(int(e + (f - e) * t) for f, e in zip(FULL_COLOR, EMPTY_COLOR))$
Shell cwd was reset to /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander
```

CLAUDE

The blank lines before the new constants were off by one. Fixing that replacement:

The blank lines before the new constants were off by one. Fixing that replacement:

Tool call: Bash

Rebuild stages, verify final match, smoke-test each stage

```
$ S=/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad && cd $S && python - <<'EOF'
final = open("final_game.py").read()
src = open("orig_game.py").read()

def seg(start, end):
    i = final.index(start); return final[i:final.index(end, i)]

def rep(s, old, new):
    assert s.count(old) == 1, old; return s.replace(old, new)

s1 = rep(src, "
            if pad and abs(self.vel.y) <= MAX_SPEED_Y and abs(angle) <= MAX_ANGLE:\n",
            "
            if (pad and abs(self.vel.x) <= MAX_SPEED_X and abs(self.vel.y) <= MAX_SPEED_Y\n
and abs(angle) <= MAX_ANGLE):\n")
s1 = rep(s1, '
            reason = "bad angle"\n
            else:',
            '
            reason = "bad angle"\n
            elif abs(self.vel.x) > MAX_SPEED_X:\n
reason = "drifting sideways"\n
            else:')

s2 = rep(s1, "\n\n\ndef ship_color(fuel_ratio):", "\n\n\nFULL_COLOR, EMPTY_COLOR = (230, 230, 240),
(255, 50, 40)\n\n\ndef ship_color(fuel_ratio):")
s2 = rep(s2, '""Return an (r, g, b) hull colour for the given fuel ratio (1.0 = full), or None for the
default.""\n
    pass',
    seg('""Return an (r, g, b) hull colour', "\n\n\ndef on_landing"))

s3 = rep(s2, "(255, 50, 40)\n", "(255, 50, 40)\n" + seg("FIREWORK_COLORS", "\n\n\ndef ship_color") +
"\n")
s3 = rep(s3, '""Called after a successful landing with the points just earned; add fireworks or
bonuses here.""\n
    pass',
```

```

        seg('""""Called after a successful landing', "\n\n\ndef bonus_life_threshold"))
s3 = rep(s3, "(WIDTH // 2, HEIGHT // 3))\n", "(WIDTH // 2, HEIGHT // 3))\n
draw_particles(screen)\n")
s3 = rep(s3, "            game.update(dt, pygame.key.get_pressed())\n", "            game.update(dt,
pygame.key.get_pressed())\n            update_particles(dt)\n")

s4 = rep(s3, '""""Return a score value at which the player earns an extra life, or None to disable bonus
lives.""""\n    pass',
        '""""Return a score value at which the player earns an extra life, or None to disable bonus
lives.""""\n    return 1500')

assert s4 == final, "stage 4 does not reproduce the final file"
for n, s in enumerate([src, s1, s2, s3, s4]):
    open(f"stage{n}.py", "w").write(s)
    compile(s, f"stage{n}", "exec")
print("all stages built; stage4 == final")
EOF
for i in 1 2 3 4; do echo "== stage $i: $(diff stage$((i-1)).py stage$i.py | grep -c '^[<>]') changed
lines"; done
for i in 1 2 3 4; do SDL_VIDEODRIVER=dummy timeout 3 python -c "
import importlib.util,pygame
spec=importlib.util.spec_from_file_location('g','stage$i.py'); g=importlib.util.module_from_spec(spec);
spec.loader.exec_module(g)
pygame.init(); s=pygame.display.set_mode((800,600)); G=g.Game();
G.update(0.016,pygame.key.get_pressed()); G.draw(s); print('stage $i runs')" 2>/dev/null | tail -1;
done

```

Result

```

all stages built; stage4 == final
== stage 1: 5 changed lines
== stage 2: 6 changed lines
== stage 3: 33 changed lines
== stage 4: 2 changed lines
stage 1 runs
stage 2 runs
stage 3 runs
stage 4 runs
Shell cwd was reset to /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander

```

CLAUDE

Every stage imports, updates and draws without errors, and stage 4 is byte-identical to the tested final file. Now the commits, staging only the lab files by path (the handout PDF stays untracked):

Tool call: Bash

Create baseline commit plus one commit per task

```

$ S=/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-
1630c17fdf4a/scratchpad
R=/home/aditya/Documents/SE/SE-LABS; L=Lab-04/32_lunarLander; CO="Co-Authored-By: Claude Opus 5.5
<noreply@anthropic.com>"
set -e
cd $R
cp $S/stage0.py $L/game.py
git add $L/game.py $L/README.md $L/.gitignore
git commit -q -m "Add Lab-04 Lunar Lander starter code" -m "Imported from SETAPESU26/32_lunarLander
(unmodified baseline)."
```

```

cp $S/stage1.py $L/game.py; git add $L/game.py
git commit -q -m "Task 1: Check horizontal speed in touchdown validation" -m "touchdown() never
compared vel.x against MAX_SPEED_X, so a sideways slam onto a pad at safe vertical speed and angle

```

```
still counted as a perfect landing. Also report 'drifting sideways' as the crash reason.
```

```
$C0"
cp $S/stage2.py $L/game.py; git add $L/game.py
git commit -q -m "Task 2: Implement ship_color to shift hull toward red as fuel runs low" -m "Linearly
blends the default hull colour (230, 230, 240) at full fuel to red (255, 50, 40) at empty.

$C0"
cp $S/stage3.py $L/game.py; git add $L/game.py
git commit -q -m "Task 3: Implement on_landing fireworks effect" -m "Spawns staggered particle bursts
after a successful landing; burst count scales with points earned, so the x3 pad gets a bigger show.
Particles update from the main loop because Game.update() stops running once the ship has landed.

$C0"
cp $S/stage4.py $L/game.py; git add $L/game.py
git commit -q -m "Task 4: Award a bonus life every 1500 points" -m "$C0"
git log --oneline -6 && git status --short && cmp $L/game.py $S/final_game.py && echo "working tree
game.py == tested final"
```

Result

```
755b519 Task 4: Award a bonus life every 1500 points
a15b494 Task 3: Implement on_landing fireworks effect
ba105a9 Task 2: Implement ship_color to shift hull toward red as fuel runs low
d2c70b5 Task 1: Check horizontal speed in touchdown validation
8541e23 Add Lab-04 Lunar Lander starter code
4974344 Finish lab 3 deliverables
?? Lab-04/Lab_4_VibeCoding_Student_handout.pdf
working tree game.py == tested final
Shell cwd was reset to /home/aditya/Documents/SE/SE-LABS/Lab-04/32_lunarLander
```

Tool call: Bash

Find how the mid-turn user message is stored

```
$ grep -n "create 1 commit for each task" ~/.claude/projects/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a.jsonl | cut -c1-600 | head -3
```

Result

```
145:{"type":"queue-operation","operation":"enqueue","timestamp":"2026-09-30T03:16:49.609Z","sessionId":"b52081dc-61a0-4d82-bf2d-1630c17fdf4a","content":"create 1 commit for each task"}
146:{"type":"queue-operation","operation":"remove","timestamp":"2026-09-30T03:16:51.359Z","sessionId":"b52081dc-61a0-4d82-bf2d-1630c17fdf4a","content":"create 1 commit for each task","reason":"absorbed_mid_turn"}
149:{"parentUuid":"921b8c74-54a9-47d3-9aef-529ef8f25eda","isSidechain":false,"attachment":{"type":"queued_command","prompt":"create 1 commit for each task","source_uuid":"0ad6f0d1-85c9-42da-ae6a-ce39aad9ef2e","commandMode":"prompt","origin":{"kind":"human"},"timestamp":"2026-09-30T03:16:49.609Z","humanTurn":true},"type":"attachment","uuid":"62f01762-c6bc-4f45-84d1-81a790d9860c","timestamp":"2026-09-30T03:16:49.609Z","rendered":[{"content":"<system-reminder>\n\nThe user sent a new message while you were working:\n\ncreate 1 commit for each task\n\n\nThis is how Claude Code surfaces messages the user
```

CLAUDE

It's stored as a `queued\_command` attachment, which the exporter currently skips. I'll add it:

Tool call: Edit

/tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/export_chat.py

```
d = json.loads(line)
if d.get("type") not in ("user", "assistant") or d.get("isSidechain"):
```

```
d = json.loads(line)
att = d.get("attachment") or {}
if d.get("type") == "attachment" and att.get("type") == "queued_command" and att.get("humanTurn"):
    rows.append(f"<div class='msg user'><div class=who>User (sent mid-task)</div><div class=text>
{esc(att['prompt'])}</div></div>")
    continue
if d.get("type") not in ("user", "assistant") or d.get("isSidechain"):
```

Result

The file /tmp/claude-1000/-home-aditya-Documents-SE-SE-LABS-Lab-04-32-lunarLander/b52081dc-61a0-4d82-bf2d-1630c17fdf4a/scratchpad/export_chat.py has been updated successfully. (file state is current in your context – no need to Read it back)